

WASP-2b

R-1426

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_220802 · created 2026-08-01T06:54:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459793.898 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.016
Transit depth (Rp/Rs)^2	1.28 +/- 0.36 [%]
Orbital Inclination (inc)	86.63 +/- 1.65
Airmass coefficient 1 (a1)	1.324 +/- 0.013
Airmass coefficient 2 (a2)	-0.1586 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	0.95 %
Best Comparison Star	None
Optimal Aperture	3.32
Optimal Annulus	14.67
Transit Duration (day)	0.083 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.016 vs 0.1326 (deviation 1.02σ)
Duration fit vs published	0.083 d vs 0.0737 d (deviation 0.59σ)
Mid-time O–C	-11.2 min
Depth SNR	5.9
Residual scatter	0.95 %

Light curve

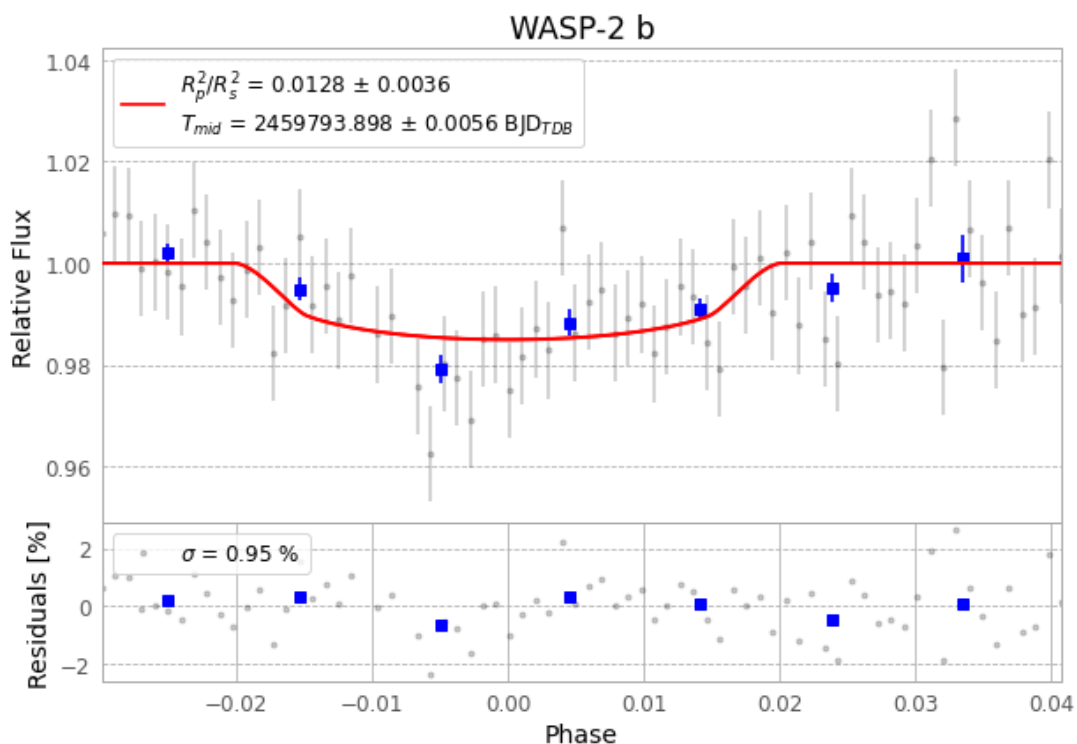


Figure 1 — FinalLightCurve_WASP-2 b_2022-08-02.png (SHA-256 127cd70eccc32880...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 26ec30dbbafb3482...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-2220802075113.FITS → WASP-2220802113013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-220802.log (b7d8172a88bb...), FinalLightCurve_WASP-2 b_2022-08-02.png (127cd70eccc3...), FinalLightCurve_WASP-2 b_2022-08-02.csv (dd77775bf848...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:54Z.